

HOME NETWORK TRAFFIC MONITOR

Architecture Design Document
No Raspberry Required Edition

Version 1.1
Status Draft
Date May 2026

Main Recommendation

Use an existing always-on computer as the monitoring host. Run Pi-hole or AdGuard Home for DNS visibility, mitmproxy for opted-in HTTPS inspection on devices you own, and tshark for local packet metadata when the network topology allows it. This avoids buying a Raspberry Pi, but it changes the capture model: full network interception requires a router feature, an inline gateway, or port mirroring.

1. Overview

This document defines the architecture for a home network traffic monitoring system designed to capture, store, analyze, and visualize traffic from devices on a residential network. The system is optimized for defensive monitoring, malware investigation, and visibility into suspicious outbound connections.

The original design assumed a Raspberry Pi as the dedicated capture device. This revised design removes that requirement and uses existing hardware whenever possible: a desktop PC, spare notebook, mini PC, NAS, home server, or compatible router.

Core Reality

You do not need a Raspberry Pi. You do need some device or router feature that can observe, route, proxy, or log traffic. Without a capture point, a normal computer cannot magically see all Wi-Fi traffic from every device.

1.1 Objectives

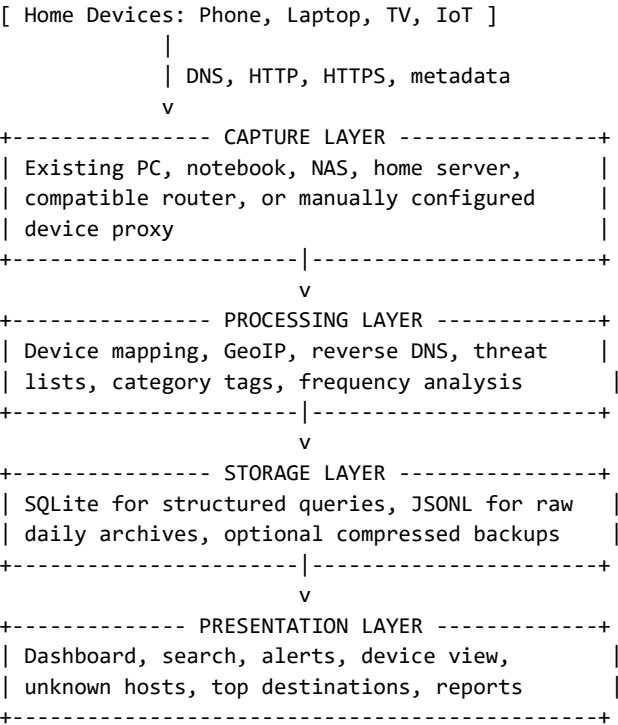
- Log DNS requests from all devices when the router can force or advertise a local DNS server
- Capture full HTTP and inspect HTTPS for devices that you own and explicitly configure
- Associate events with source device using IP address, MAC address, DHCP name, or manual device mapping
- Store logs locally in a searchable format for forensic review
- Provide a dashboard for real-time and historical traffic inspection
- Alert on suspicious patterns such as unknown destinations, high-frequency calls, unusual countries, and large outbound payloads
- Support an investigation workflow that helps identify suspicious apps and remove or block them

1.2 Scope

In Scope	Out of Scope
Home Wi-Fi and wired devices	Traffic on 4G or 5G cellular networks
DNS visibility for most devices	Breaking certificate pinning in banking or payment apps
HTTP traffic and HTTPS from opted-in devices	Intercepting traffic from guests or other people without explicit consent
IoT metadata through DNS, packet metadata, or router logs	Guaranteeing full payload visibility from all IoT devices
Local dashboard, local logs, and local alerts	Enterprise SIEM replacement or corporate compliance monitoring

2. Architecture Strategy Without Raspberry

The system has the same logical layers as the original architecture: Capture, Processing, Storage, and Presentation. The difference is where the Capture layer runs. Instead of a Raspberry Pi, the Capture layer can run on an existing machine or router.



2.1 Recommended No-Purchase Architecture

Layer	Recommended Implementation	Why
Capture	Pi-hole or AdGuard Home on an existing always-on computer; mitmproxy for opted-in devices; router logs if available	Gives useful visibility without a new device
Processing	Local Python service that reads DNS logs, proxy logs, and optional packet metadata	Keeps enrichment logic simple and local
Storage	SQLite for queries plus JSONL for daily raw archives	Searchable, portable, easy to back up
Presentation	Local web dashboard using FastAPI plus React, or Grafana if you prefer faster setup	Enough for live feed, filters, charts, and alerts
Alerts	Telegram bot, email SMTP, or local dashboard notifications	Simple and practical for home monitoring

3. Network Topology Options

The topology determines what the system can actually see. This is the most important design decision. The less control you have over the router, the less complete the capture will be.

Option	Hardware Needed	Visibility	Best For	Tradeoff
A. DNS-only monitor	Existing PC or NAS running Pi-hole or AdGuard Home	Domains queried by each device	Fastest no-purchase setup	No request body, no URL path, no HTTPS payload
B. Opt-in HTTPS proxy	Existing PC running mitmproxy	Full HTTP and HTTPS for devices you manually configure	Your laptop, phone, test devices	Each device must trust the proxy CA and use the proxy
C. Router-integrated capture	Compatible router with OpenWrt, tcpdump, DNS logs, or firewall rules	DNS, firewall events, and sometimes packet metadata	Networks where the router is flexible	Depends heavily on router support
D. Existing PC as gateway	Old notebook or desktop with two interfaces, or Ethernet plus Wi-Fi hotspot	Most complete network-wide routing visibility	Advanced setup without Raspberry	The PC must stay on and may become a bottleneck
E. Managed switch port mirror	Only if you already own a managed switch	Read-only packet metadata from mirrored traffic	IoT metadata and passive observation	No blocking and no HTTPS decryption

Best Practical Choice

Start with Option A plus Option B. Use DNS monitoring for every device and HTTPS proxying only for devices you own and actively want to inspect. This gives strong malware visibility without needing to redesign the entire network.

4. Component Descriptions

4.1 Capture Layer

The Capture layer collects raw events. In the no-Raspberry design, capture is split into three channels: DNS logs, proxy logs, and optional packet metadata.

Capture Channel	Tool	What It Captures	Works Without Raspberry	Notes
DNS	Pi-hole or AdGuard Home	Domain lookups by device	Yes	Best baseline for all devices
HTTP and HTTPS proxy	mitmproxy	Full requests and responses for opted-in devices	Yes	Requires local CA on your own devices
Packet metadata	tshark or tcpdump	IPs, ports, protocols, timing, packet sizes	Yes, if the machine can see the traffic	Useful for IoT and non-standard ports
Router logs	OpenWrt, UniFi, AsusWRT, MikroTik, pfSense, OPNsense	DNS, firewall, DHCP, device data, sometimes flows	Yes	Depends on router capabilities
Infrastructure monitoring	Zabbix, Nagios, LibreNMS, PRTG	Device health, latency, uptime, bandwidth	Yes	Useful as a secondary layer, not the main malware inspection layer

4.2 Why Zabbix, Nagios, PRTG, and LibreNMS Are Not the Core Stack

Traditional network monitoring tools are excellent for availability, device health, bandwidth, latency, SNMP metrics, and alerting. They are not ideal as the primary tool for inspecting home device requests, identifying suspicious domains, or reviewing app-level traffic. They can be added later for device health, but the core malware investigation stack should stay focused on DNS, proxy logs, and metadata.

4.3 Processing Layer

The Processing layer normalizes and enriches every event before it reaches storage.

Enrichment	Description
Device resolution	Maps source IP or MAC address to a friendly device name using DHCP leases, router data, or a manual device inventory
Reverse DNS	Resolves destination IP addresses when hostname is missing
GeoIP lookup	Adds country and city metadata from a local GeoIP database
Threat intelligence tag	Checks destination IP or domain against local blocklists and suspicious host feeds
Category tag	Classifies destination as ads, analytics, CDN, social, unknown, software update, banking, or messaging
Frequency analysis	Counts repeated calls to the same destination per device and per time window
Timing and size	Records request duration, payload size, response size, and time of day
Privacy filter	Redacts Authorization, Cookie, Set-Cookie, API keys, and selected query parameters before storage

4.4 Storage Layer

Store	Purpose	Retention
SQLite	Structured queries for dashboard filters, device views, alerts, top destinations, and unknown hosts	30 days by default
JSONL daily archive	Raw event archive for deeper forensic review and incident preservation	7 to 30 days by default, configurable
Compressed archive	Gzip files for older JSONL logs	Optional, based on disk space
Device inventory file	Friendly names, MAC addresses, owner, device type, and expected destinations	Persistent
Alert history	Confirmed suspicious events and investigation notes	Persistent or 90 days

4.5 Presentation Layer

View	Purpose
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Live Feed	Shows captured events as they arrive
Device Overview	Shows traffic volume, top domains, categories, and unusual activity per device
Unknown Hosts	Highlights domains not categorized or not expected for that device
Threat Alerts	Lists matches from threat feeds and local blocklists
Top Destinations	Ranks most contacted hosts per time window
Geo Map	Shows countries receiving traffic and flags unusual regions
Search	Filters by device, host, URL, method, status, category, threat flag, size, and time
Incident Notes	Stores investigation status, confirmed cause, action taken, and evidence link

5. Data Flow

Because this architecture avoids buying a Raspberry Pi, the system has two primary flows: DNS-first monitoring for the whole home and opt-in HTTPS inspection for selected devices.

5.1 DNS Monitoring Flow

1. A device connects to the home router and receives the local DNS server through DHCP, or the router forwards DNS to the monitoring host.
2. The device requests a domain such as api.example.com.
3. Pi-hole or AdGuard Home logs the query, source IP, timestamp, and response.
4. The Processing Agent reads the DNS event and maps the source IP to a friendly device name.
5. The event receives category, GeoIP, reverse DNS, and threat intelligence enrichment.
6. The enriched DNS event is stored in SQLite and appended to the daily JSONL archive.
7. If the domain is suspicious, unknown, newly seen, or high-frequency, an alert is generated.
8. The dashboard updates the device timeline, top destinations, unknown hosts, and alert views.

5.2 Opt-in HTTPS Inspection Flow

9. The user configures their own phone or laptop to use the monitoring host as an HTTP and HTTPS proxy.
10. The user installs the local CA certificate only on devices they own and consent to inspect.
11. The device makes a request to a website or API.
12. mitmproxy terminates TLS locally, reads the plaintext request, and forwards it securely to the destination.
13. A mitmproxy addon exports request and response metadata to the Processing Agent.
14. Sensitive headers and tokens are redacted before storage unless deep forensic mode is explicitly enabled.
15. The enriched event is stored in SQLite and JSONL.
16. The dashboard shows URL, method, host, status, size, timing, category, and threat status.
17. If a suspicious destination or payload pattern appears, the alert workflow begins.

5.3 Optional Packet Metadata Flow

18. tshark or tcpdump runs on a machine that can see traffic: the router, an inline gateway, a mirrored switch port, or the monitored machine itself.
19. The tool records metadata such as source IP, destination IP, port, protocol, byte counts, and timing.
20. The Processing Agent correlates metadata with DNS logs and device identity.
21. The system flags non-standard ports, unusual destinations, unexplained high-volume outbound traffic, and odd-hour activity.
22. Metadata is stored without decrypting payloads.

6. Deployment Guide Without Raspberry

6.1 Hardware You Can Use Instead

Existing Device	Good For	Important Requirement
Desktop PC	Running Docker services, dashboard, Pi-hole, mitmproxy, and storage	Must stay on when you want monitoring active
Old notebook	Dedicated monitoring host or temporary gateway	Prefer Ethernet for stability
Mac mini or mini PC	Quiet always-on monitoring server	Best no-Raspberry replacement if you already own one
NAS	Running DNS and dashboard containers	Needs Docker or app support
Router with OpenWrt, pfSense, OPNsense, MikroTik, UniFi, or similar	DNS enforcement, firewall logs, DHCP inventory, optional packet capture	Router must expose logs or support packages

6.2 Recommended Setup Path

23. Choose one always-on host: desktop, notebook, NAS, mini PC, or compatible router.
24. Give the host a static IP address on the local network.
25. Run Pi-hole or AdGuard Home on that host for DNS visibility.
26. Configure the router DHCP settings to advertise the monitoring host as the DNS server. If the router does not allow this, manually set DNS on priority devices.
27. Run mitmproxy on the same host for selected devices you own and want to inspect deeply.
28. Install the local CA certificate only on those selected devices.
29. Create a device inventory with friendly names, IP addresses, MAC addresses, and expected apps or services.
30. Run the Processing Agent to normalize DNS logs, proxy logs, and optional packet metadata.
31. Store enriched events in SQLite and JSONL.
32. Open the local dashboard and verify that device traffic appears correctly.
33. Configure alerts for threat intelligence matches, unknown domains, high-frequency calls, large outbound payloads, and unusual countries.
34. Review logs for 24 to 48 hours to build a normal baseline before treating unknown traffic as malicious.

6.3 Minimum Viable Setup

Minimum Viable Setup

One existing always-on computer, Pi-hole or AdGuard Home, a static local IP address, router DNS pointing to that host, SQLite storage, and a basic dashboard. This is enough to see which domains each device contacts and to start detecting suspicious DNS behavior.

6.4 Full Visibility Setup Without Raspberry

For near-complete network visibility without buying a Raspberry Pi, use an existing old notebook or desktop as the gateway. This requires two network paths: one side connected to the router and one side serving devices through Ethernet, switch, or Wi-Fi hotspot. This setup is more complex, but it is the closest replacement for the original Raspberry design.

- Use the existing computer as the default gateway for monitored devices
- Run DNS monitoring, mitmproxy, packet metadata capture, processing, storage, and dashboard on the same host
- Keep a fallback plan because if this host shuts down, monitored devices can lose internet access
- Avoid this mode if you need a simple, low-maintenance home setup

7. Security and Privacy Considerations

Security Principle

The monitoring host becomes sensitive infrastructure. It may store DNS history, URLs, headers, cookies, tokens, device identities, and personal browsing metadata. Treat it like a security asset.

7.1 Host Hardening

- Use a dedicated local user for the monitoring services
- Disable remote access unless it is needed
- Use SSH keys instead of passwords when SSH is enabled
- Restrict dashboard access to the local network or through Tailscale or a private VPN
- Keep the operating system and packages updated
- Encrypt the disk when logs are stored on a portable notebook or external drive
- Do not expose the dashboard directly to the public internet

7.2 CA Certificate Management

- Use HTTPS inspection only on devices you own and consent to inspect
- Store the CA private key with restricted permissions
- Rotate the CA certificate if the monitoring host is compromised
- Expect certificate pinning to block inspection in banking, payment, and security apps

- Keep an exclusion list for apps that should never be inspected

7.3 Log Protection

- Redact Authorization, Cookie, Set-Cookie, and API key values before storage by default
- Limit log retention to what you actually need
- Separate raw archives from dashboard data
- Preserve only relevant JSONL files during an incident
- Do not collect traffic from guests, shared users, or family members without clear permission

8. Malware Detection Workflow

8.1 Suspicious Indicators

Indicator	What To Look For	Likely Source
Unknown destination	Domains or IPs not associated with any installed app, service, update system, or known vendor	DNS logs, proxy logs
High-frequency calls	Hundreds of requests per hour to the same endpoint	DNS logs, proxy logs, metadata
Large outbound payloads	Large POST or upload activity to unknown hosts	Proxy logs, metadata
Odd-hour traffic	Heavy traffic late at night when nobody is using the device	All logs
Non-standard ports	Connections outside expected ports such as 80 and 443	Packet metadata, router logs
Threat intelligence match	Destination appears in local blocklists or suspicious host feeds	Processing layer
Geo anomaly	Background traffic to countries or regions that are unusual for the household	GeoIP enrichment
DNS without follow-up	Repeated DNS queries without normal traffic patterns	DNS logs and metadata

8.2 Investigation Steps

35. Open the dashboard and go to Unknown Hosts or Threat Alerts.
36. Filter by the suspect device.
37. Review destination host, destination IP, frequency, request size, response size, time pattern, and country.
38. Check whether the destination matches a recently installed app, browser extension, IoT device, or background service.
39. Search the destination in trusted reputation tools from a separate browser session.
40. If the traffic is suspicious but not confirmed, block the domain at the DNS level temporarily and watch what breaks.
41. If confirmed malicious, remove the app or isolate the device.
42. Reset credentials if sensitive data may have been sent.
43. Preserve the relevant JSONL archive before the retention period deletes it.
44. Document the incident with device, destination, evidence, action taken, and result.

9. Known Limitations

Limitation	Explanation	Mitigation
No capture point	If no router feature, gateway, proxy configuration, or mirrored traffic exists, you cannot see everything from a normal computer.	Start with DNS monitoring and opt-in proxying
Certificate pinning	Some apps reject local MITM certificates by design.	Exclude those apps and rely on DNS or metadata
DNS over HTTPS or DNS over TLS	Devices may bypass local DNS logging.	Block common encrypted DNS endpoints or enforce router DNS where possible
QUIC and HTTP/3	Some HTTPS traffic uses UDP 443 and may bypass proxy behavior.	Block UDP 443 to force TCP fallback when appropriate
VPN traffic	A VPN hides the internal destination and payload from the local monitor.	Detect VPN usage but do not expect content visibility
IoT devices	Many IoT devices cannot install CA certificates.	Use DNS logs, metadata, and VLAN isolation
Computer uptime	If the existing PC is off, DNS or proxy monitoring stops.	Use a machine that is normally always on
Performance	A PC gateway or proxy can become a bottleneck if underpowered.	Use DNS-only mode for broad coverage and proxy only selected devices

10. Future Enhancements

- Baseline learning per device to detect unusual changes in normal traffic
- Automatic device fingerprinting from DHCP names, user agents, and traffic patterns
- Local allowlist per device type, such as phone, laptop, TV, camera, or speaker
- One-click incident export with related DNS, proxy, metadata, and alert records
- Automatic DNS block suggestions for confirmed malicious destinations
- Separate IoT VLAN recommendations based on suspicious or noisy behavior
- Encrypted archive backup to local NAS or external storage

Appendix A: Glossary

Term	Definition
CA Certificate	Certificate authority certificate used by a local proxy so trusted devices can inspect HTTPS traffic
C2	Command and control server used by malware to receive instructions or send data
DHCP	Protocol that assigns local IP addresses and network settings to devices
DNS	System that translates domain names into IP addresses
GeoIP	Database that maps IP ranges to countries and cities
JSONL	JSON Lines format where each line is a separate JSON record
MITM	Man in the middle proxy that inspects traffic between a client and destination with consent
Pi-hole	Open-source DNS sinkhole used to block or log domains at the network level
tshark	Command-line packet analysis tool based on Wireshark